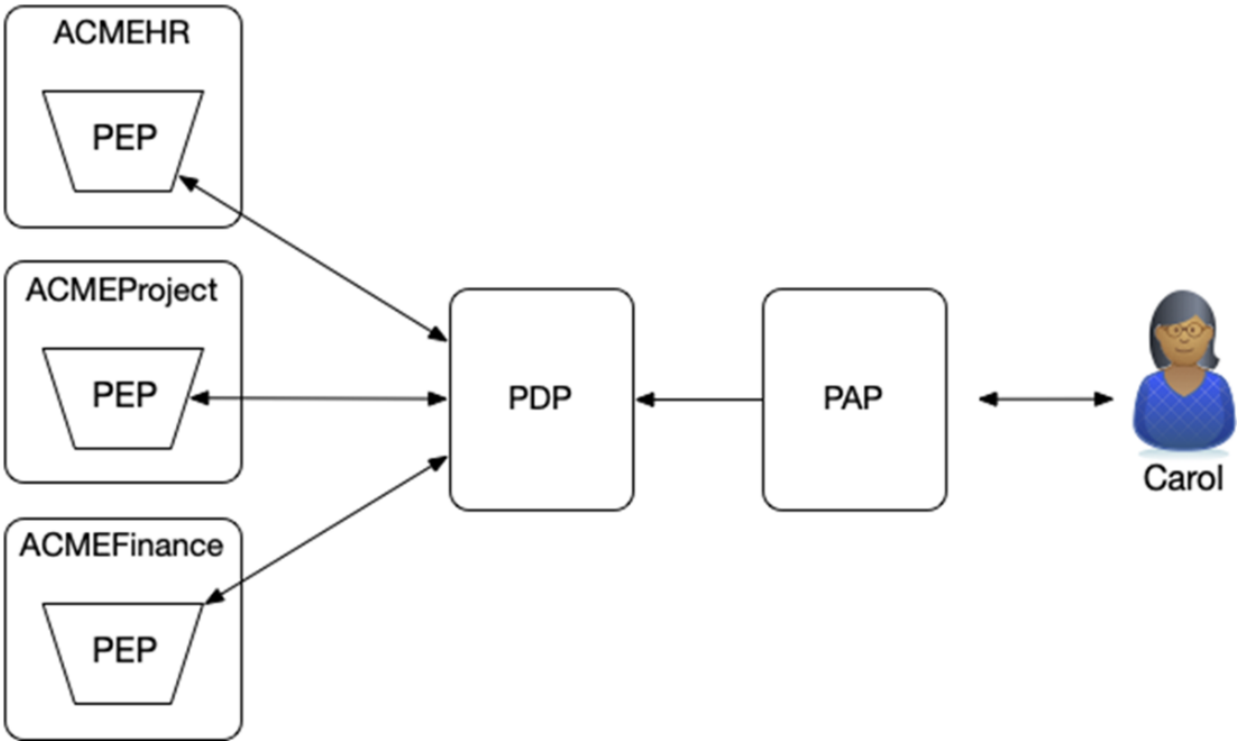


Figure 4.6 Centralized authorization architecture at ACME Corp. Multiple application-specific PEPs send access requests to a shared PDP, which evaluates them using policies authored and managed by Carol in the PAP. This approach centralizes decision-making and simplifies policy updates, but it introduces potential availability concerns.

This centralized architecture offers a single source of truth for authorization logic. When Carol, ACME's security administrator, updates a policy via the Policy Administration Point (PAP), the PDP immediately begins using the new rules to evaluate access requests across all connected applications. This streamlines policy management, enhances consistency, and centralizes logging and auditing.

However, there's a trade-off. Since all PEPs rely on the same PDP, any latency, failure, or unavailability in the decision service can degrade the performance of all dependent systems. In high-scale or latency-sensitive environments, this dependency might be a concern, requiring careful design for availability, caching, or fallback strategies.

4.5.3 Distributing authorization decisions

As ACME Corp's systems become more distributed, the limitations of a centralized PDP may become more pronounced. Applications that require